

Name: **Group 80**

Student Reference Number:

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Deadline Date: 9th May 2025	Member of staff responsible for coursework: Dr Taimur Bakhshi
Programme:	
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Group work: please list all names of all participants formally associated with this work and state whether the work was undertaken alone or as part of a team. Please note you may be required to identify individual responsibility for component parts.	
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Roles and Responsibilities

Student id	Role	Responsibilities
Udagedara Premachandre	Project Manager	Task distribution, and coordinated team communication.
Bamunu Kumaradasa	Frontend Developer	Developed UI using React and Chakra UI, implemented routing and responsiveness.
Kodippili Prabhathiya	3D Model Integrator	Handled integration of Babylon.js models, lighting, and shadows.
Wanigasooriya Wanigasooriya	UI/UX Designer	Designed the interface layout, and color themes.
Gamlath Gamlath	Frontend Developer	Developed UI using React and Chakra UI, implemented routing and responsiveness.
Mallawa Mallawathanthri	Testing & Evaluation	Conducted usability testing, managed surveys, and analyzed user feedback for updates.

Project Links

GitHub repository (source code):

<https://github.com/Nipun-Malinga/furniture-store.git>

YouTube video presentation:

<https://youtu.be/fBlWjmgWEc>

1. Introduction

This project aims to develop an interactive web-based 3D furniture customization application using the latest JavaScript technologies such as React, Babylon.js, Chakra UI, and Zustand. The application is intended to assist interior designers and clients in co-designing furniture arrangements concerning room sizes, geometries, and colors.

With the growth of online shopping, the problem of seeing furniture in practical and structurally correct terms continues to be a top limiting factor. The application overcomes this handicap by being a browser-based application that combines real-time 3D rendering with a user-friendly and responsive interface. It enables users to choose, set, customize, and relocate pieces of furniture like sofas, chairs, beds, and tables, and modify their size and color while observing the arrangement in both 2D and 3D environments.

The application, developed with React and driven by Babylon.js for 3D visualization, includes real-time scene updating, theming in an interactive manner, and offering engaging and professional design experience. For providing an accessible and simple interface on devices based on Chakra UI. Furniture zooming, color tweaking, and room presetting were implemented using component-based design patterns and global state management through Zustand.

Development took place through an agile process beginning with requirement gathering via surveys and casual interviews. This document presents the entire project life cycle from ideation and high-fidelity prototyping to implementation, user testing, and evaluation.

1.1 Application Features

- **User login:**
There is a login mechanism supporting an authenticated designer to use this application.
- **Room Dimensions:**
Users can specify room dimensions (width, length, and height) and create their customized rooms. These inputs dynamically configure the 3D environment to reflect the intended room setup.
- **2D Furniture Layout:**
A simplified 2D layout helps designers plan the furniture arrangement before proceeding to a 3D perspective.
- **3D Room Preview:**
The application displays a 3D real-time view of the room that has been designed using Babylon.js. The furniture items can be turned, viewed from various angles, and zoomed out or in for more realism and awareness.

- Furniture Customization
The users can interactively scale furniture to match room proportions and change the color of furniture items.
- Design Management

1.2 Functional and Non-Functional Requirements

Table 1: Functional and Non-functional Requirements

Functional Requirements	Non-Functional Requirements
The user should be able to place, move, and take away 3D furniture items.	The application must be accessible using modern web browsers like Chrome and Firefox.
The user should be able to rotate and scale the furniture inside the room arrangement.	The UI should be responsive and accessible on desktop and mobile displays.
The user should also be able to change the furniture color	The design should follow HCI principles of control, feedback, and consistency.
The user should be able to navigate between 2D and 3D.	The application must respond to interactions within 1 second.

1.3 Low-Fidelity Prototype



Figure 2: Login - wireframe

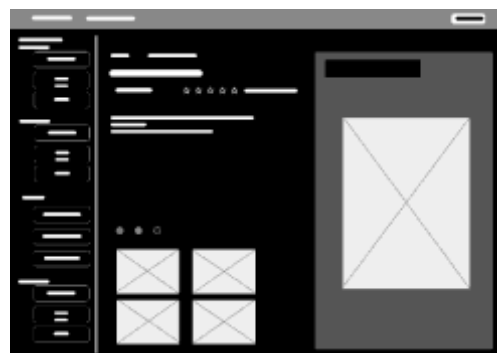


Figure 1: Furniture - wireframe

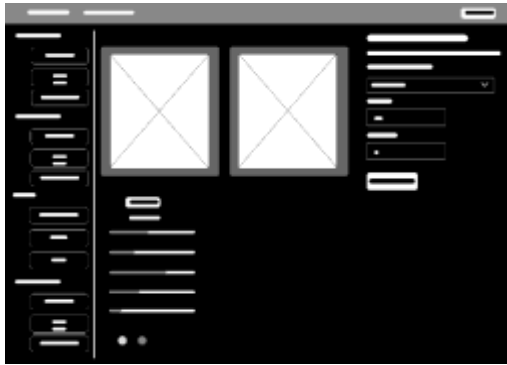


Figure 4: Room builder - wireframe



Figure 3: Room creator - wireframe 1



Figure 5: Room creator - wireframe 2

1.4 Personas and Scenarios

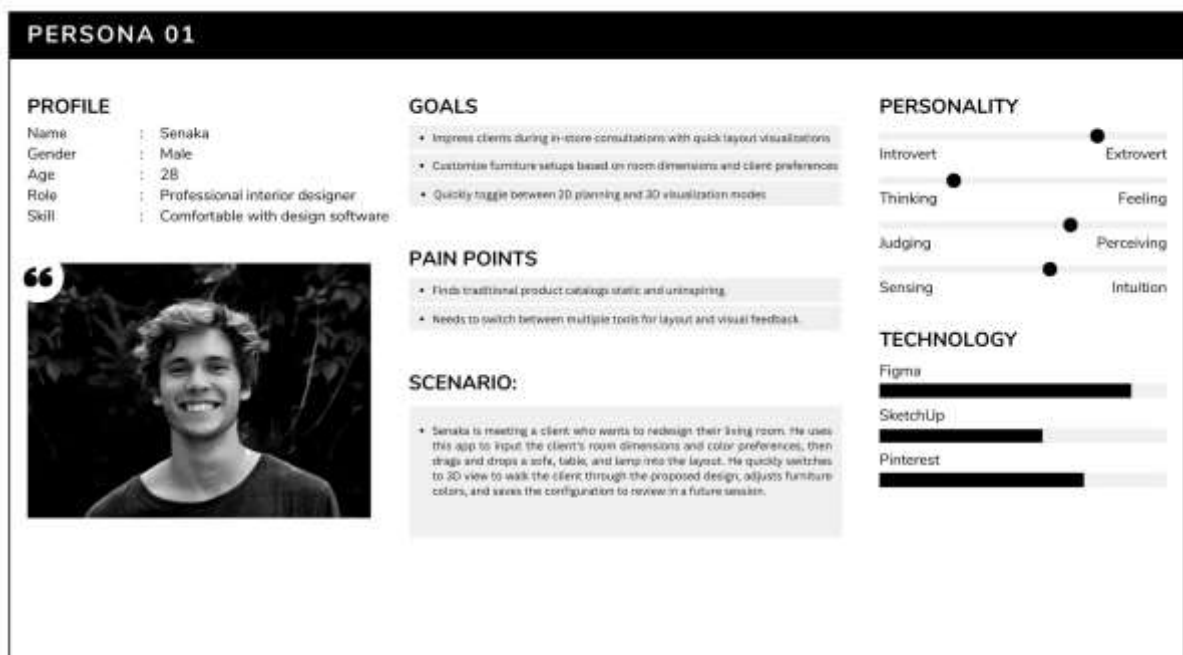


Figure 6: User persona - 1

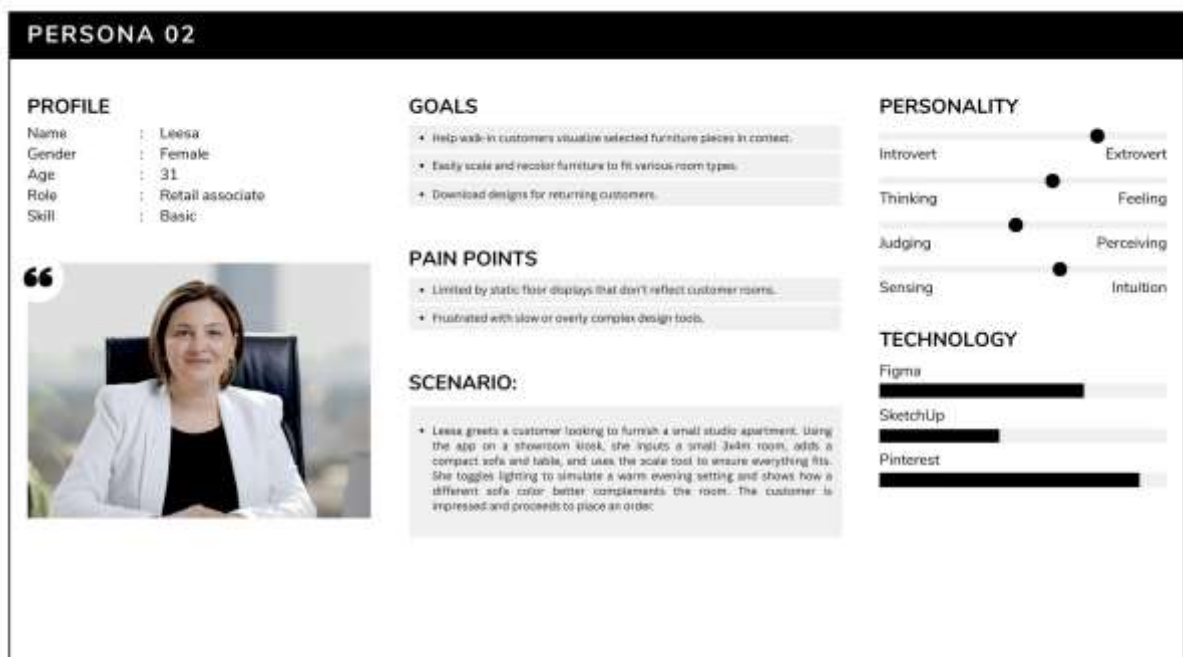


Figure 7: User persona - 2



Figure 8: User persona - 3

1.5 Storyboards



Figure 9: Storyboards

1.6 Mock Evaluations (Low-Fidelity)

A short formative user test of the app was carried out through the paper wireframe and Figma mockup to measure the usability of the low-fidelity prototype. The goal was to recognize any initial design problems with navigation, organization, and feature clarity before developing with high fidelity.

Participants:

- User 1: Interior design student, age 22
- User 2: Furniture showroom assistant, age 30
- User 3: General user with basic tech skills, age 26

Each participant was asked to complete the following tasks:

- Log in to the application
- Configure a room with custom dimensions
- Add a sofa and a table to the layout
- Switch to 3D view and inspect the design

Feedback Summary:

Table 2: Mock Evaluations Feedback

Observation	Feedback	Response
Button spacing was too close	I accidentally clicked the wrong button.	Added white space between UI controls
Icon-only buttons caused confusion	I didn't know what the cube icon was.	Added text labels instead of icons
Room arrangement fields were difficult to see	I couldn't find where to input dimensions.	Repositioned input fields to top of layout
3D toggle was intuitive and responsive	I enjoyed how simple it was to toggle between views.	Retained toggle placement and design

1.7 User Feedback (Data Gathering)

To ensure that the application is complete with user requirements and best HCI practices, we conduct structured data collection with a mixture of usability testing and web questionnaires. We found user needs, areas of pain, and design orientations towards furniture personalization and 3D room visualization.

Usability Testing Script

Participants were requested to perform primary design tasks using the Figma prototype, including:

- Adjusting the room size
- Inserting and adjusting a chair
- Previewing in 3D view

Online Survey

What is your age group?

30 responses

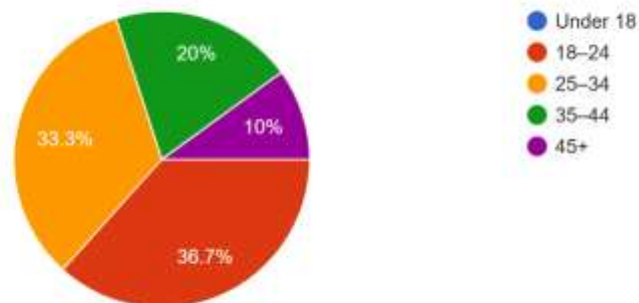


Figure 10: Survey result 1

How would you rate your experience with technology?

30 responses

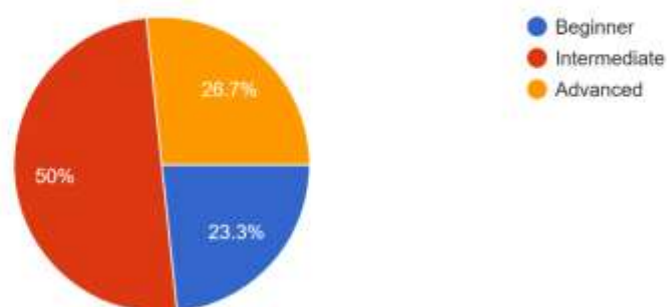


Figure 11: Survey result 2

Have you ever bought furniture online?

30 responses

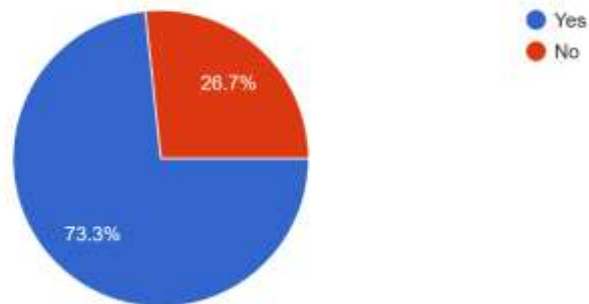


Figure 12: Survey result 3

If yes, how satisfied were you with the online visual preview of the furniture?

26 responses

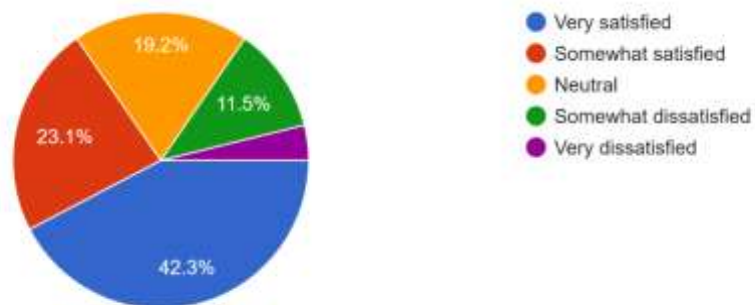


Figure 13: Survey result 4

How important is being able to view furniture in 3D before buying?

30 responses

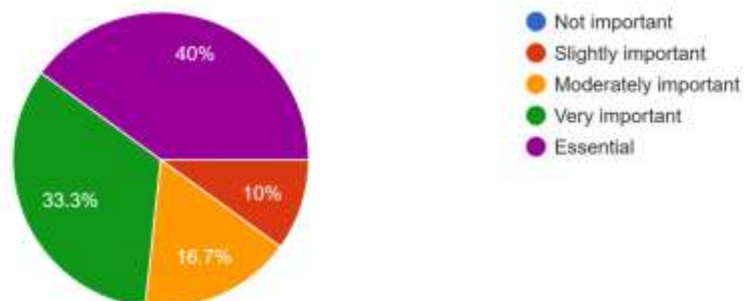


Figure 14: Survey result 5

Which features would improve your experience when designing a room? (Select all that apply)

30 responses

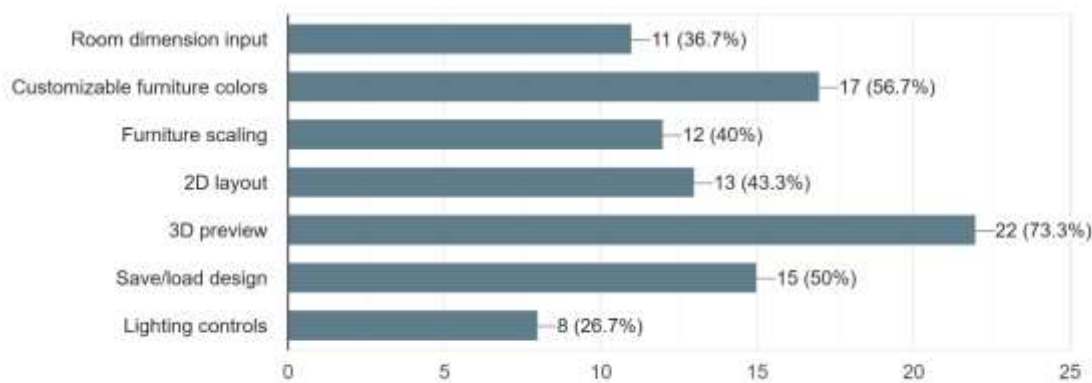


Figure 15: Survey result 6

What device would you most likely use for this kind of application?

30 responses

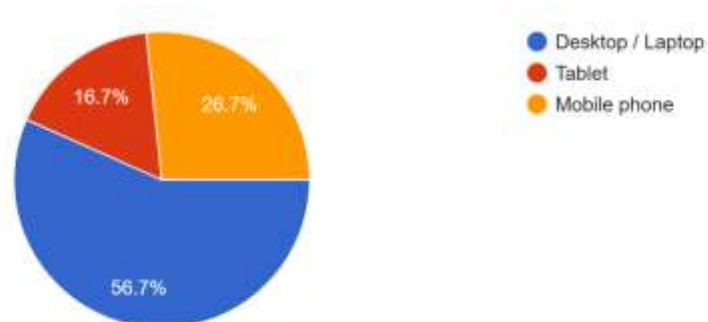


Figure 16: Survey result 7

Key findings

- 80% of users said 3D viewing is important while purchasing furniture.
- Common issues,
 - It is difficult to imagine the actual furniture size.
 - Poor understanding of navigating objects.
- Most requested functionalities
 - Room size
 - Scaling furniture
 - 3D view
 - Custom color selection
 - Save designs
- 60% of users want it to be more convenient to use the app on a mobile phone, while 40% want desktop/laptop more convenient.
- All users experienced online 3D previews are very helpful.

2. Methods and Technology

The application was developed as a web-based 3D furniture visualization platform based on the latest JavaScript libraries and frameworks. Those technologies are:

- React: To construct the user interface with component-based architecture.
- Vite: As a development tool for accelerated development and efficient bundling.
- Babylon.js: For rendering interactive 3D scenes in the browser.
- Zustand: Lightweight global state management to manage UI state, scene updates, and model configurations.
- Chakra UI: For creating a responsive and styled component library.
- React Router: For navigating between app views.

2.1 Coding Implementation

```
119     if (props.modelName) {
120       try {
121         const container = await SceneLoader.LoadAssetContainerAsync(
122           '/public/3DModels/',
123           props.modelName,
124           scene
125         );
126         container.addAllToScene();
127         modelRef.current = container.meshes[0];
128       } catch (error) {
129         console.error('Error loading model', error);
130       }
131     } else if (typeof props.model === 'function') {
132       const model = props.model(scene);
133       modelRef.current = model;
134     }
```

Figure 17: Furniture viewer - 3D model loader

```
useEffect(() => {
  if (modelRef.current && sceneRef.current && color) {
    const material = new StandardMaterial('colorMaterial', sceneRef.current);
    material.diffuseColor = Color3.FromHexString(color);

    modelRef.current.getChildMeshes().forEach((mesh) => {
      console.log(mesh.name);
      mesh.material = material;
    });
  }
}, [color, angle]);
```

Figure 18: Furniture viewer - color change


```

room &&
  rooms.items
    .filter((r) => r.value === room.selectedRoom)
    .map((r) => {
      const selectedRoom = r.room(scene, room.width, room.length);
      selectedRoom.position.y = -2;
      setSavedDesign([
        {
          modelType: 'room',
          model: {
            name: r.value,
            width: room.width,
            length: room.length,
            position: {
              x: selectedRoom.position.x,
              y: selectedRoom.position.y,
              z: selectedRoom.position.z,
            },
            scale: selectedRoom.scaling.x,
          },
        },
      ]);
    });
  });

```

Figure 19: Room builder

```

scene.onBeforeRenderObservable.add(() => {
  coordinatesRef.current.forEach(
    ({ modelId, productId, categoryId, X, Y, Z, rotation, scale, color }) => {
      const mesh = scene.getMeshByName(modelId);
      if (mesh) {
        mesh.position.set(X, Y, Z);
        mesh.rotation.y = rotation;
        mesh.scaling.setAll(scale);

        if (color) {
          const newColor = Color3.FromHexString(color);

          const applyMaterial = (targetMesh) => {
            let mat = targetMesh.material;
            if (!(mat instanceof StandardMaterial)) {
              mat = new StandardMaterial(`${targetMesh.name}-mat`, scene);
              targetMesh.material = mat;
            }
            mat.diffuseColor = newColor;
          };

          applyMaterial(mesh);

          mesh.getChildMeshes().forEach((child) => {
            applyMaterial(child);
          });
        }
      }
    }
  );
}

```

Figure 20: Room builder - furniture position logic

2.2 Evaluation & Testing

Evaluation Plan

- Method: Task-based usability testing and think-aloud protocol
- Goal: Ensure ease of use, task completion, and overall satisfaction
- Tools: Figma prototype and web app developed (React + Babylon.js)

Participants

- 5 Users
- Experience levels:
 - 2 beginners (non-tech users)
 - 2 intermediates (design/retail)
 - 1 advanced (design student)
- Type: Online

Test tasks

1. Login
2. Enter room type and dimensions
3. Add and customize furniture with color and scale
4. 3D preview
5. Save the design

Results

Table 3: Usability testing results

Task	Success	Notes
Login	100%	No issue
Room setup	60%	Inputs not clear
Add and scale furniture	80%	Some buttons not clear
Color change	100%	No issue
3D view	100%	No issue
Save design	60%	Button not easily found

Improvements

- Room setup positioned to the top.
- Make sidebar to add furniture.
- Save button enlarged.

3. Limitations

The following table shows the comparison of the core functional requirements mentioned in Appendix A with the actual status of project implementation. The percentage of completion has been estimated based on the working features available in the developed React + Babylon.js web application.

Table 4: Project criteria comparison

Objective	Completion	Comments
Customer can provide the size, shape and color scheme for the room	100%	Customer can select the width, height, and color of the room
Customer can create a new design based on room size, shape and color scheme	100%	Users can set parameters and add furniture
Customer can visualize the design in 2D	100%	Users can see 2D and 3D view side by side
Customer can visualize the design in 3D	100%	3D scene view using Babylon.js is fully working
Customers can scale the design to best fit the room	100%	Furniture can be resized using scale slides
Customer can add shade to the design as a whole or selected parts	100%	Shades are added to the whole design
Customer can change the color of the design as a whole or selected parts	100%	Users can change the color as a selected part
Customer can edit/delete the design	100%	Users can edit/delete the furniture items
Customer can save the design	100%	Users can save the design in the local storage but it not permanent

References

Shneiderman, B. and Plaisant, C. (2010). *Designing the User Interface: Strategies for Effective Human-Computer Interaction*. 5th ed. Pearson Education.

Babylon.js (2024). *Babylon.js Documentation*. [online] Available at: <https://doc.babylonjs.com> [Accessed 5 May 2025].

ReactJS (2024). *React – A JavaScript library for building user interfaces*. [online] Available at: <https://reactjs.org> [Accessed 5 May 2025].

Zustand (2024). *Zustand Documentation*. [online] Available at: <https://docs.pmnd.rs/zustand/getting-started/introduction> [Accessed 5 May 2025].

Chakra UI (2024). *Chakra UI – Build Accessible React Apps with Speed*. [online] Available at: <https://chakra-ui.com/> [Accessed 5 May 2025].

Appendix

Figma design:

<https://www.figma.com/design/9NP7XZMC9ptivF1zLlsmGt/Furniture-E-commerce--Community-?node-id=418-6164&p=f&t=DuWnIlNwDZpKxscU-0>

Data gathering form:

<https://forms.gle/9ZdUXwouz2pPhE7x8>

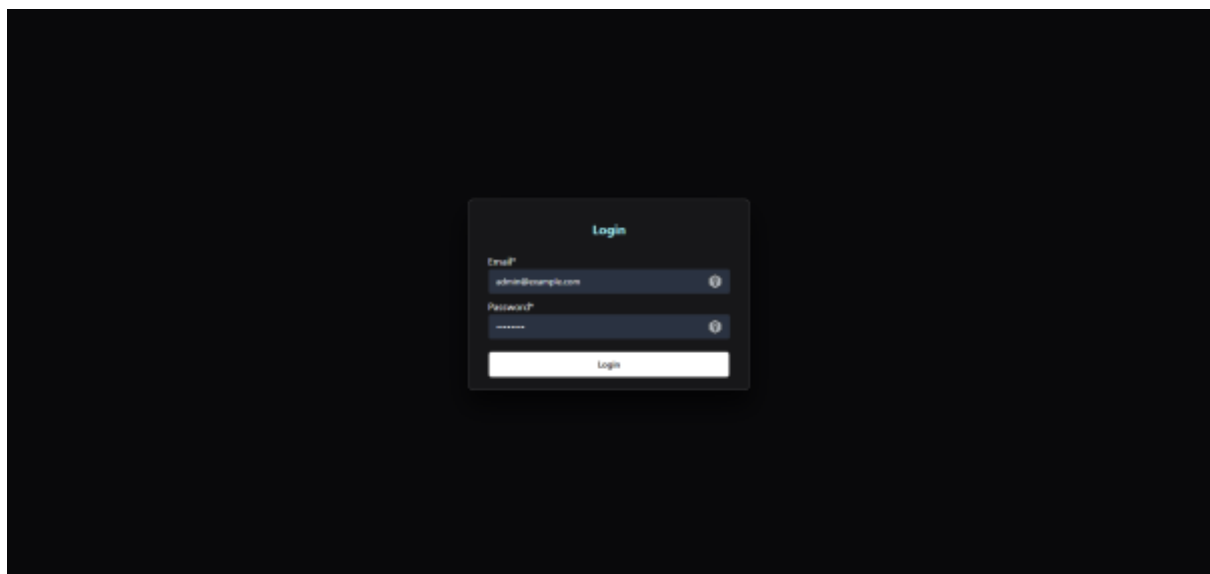


Figure 21 Login Page



Figure 22 Home Page Dark Mode



Figure 23 Home Page Light Mode

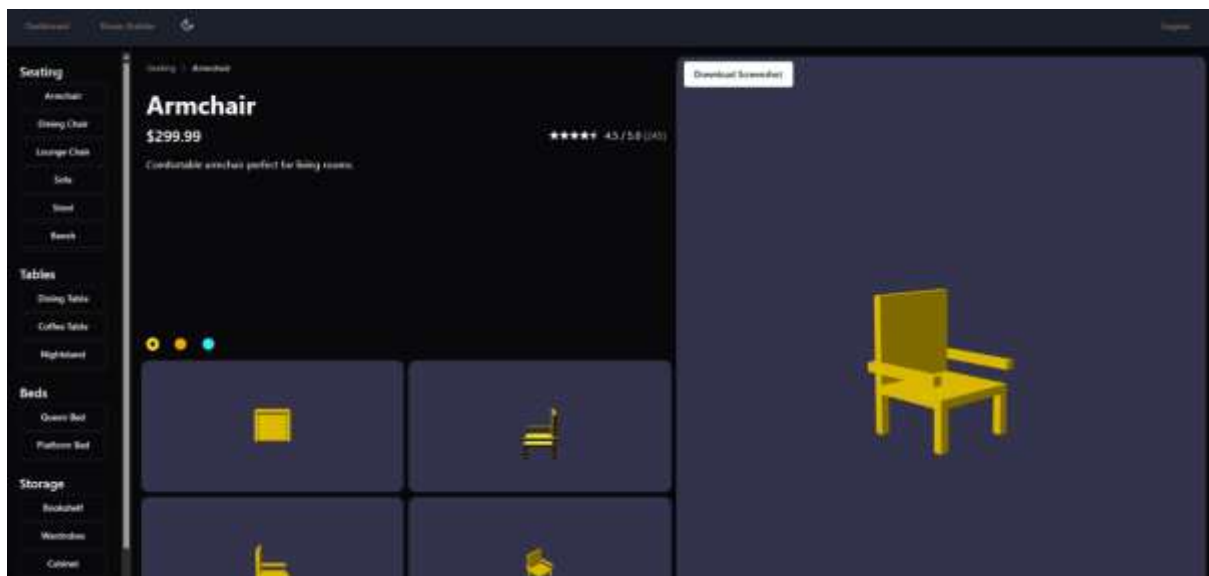


Figure 24 Home Page

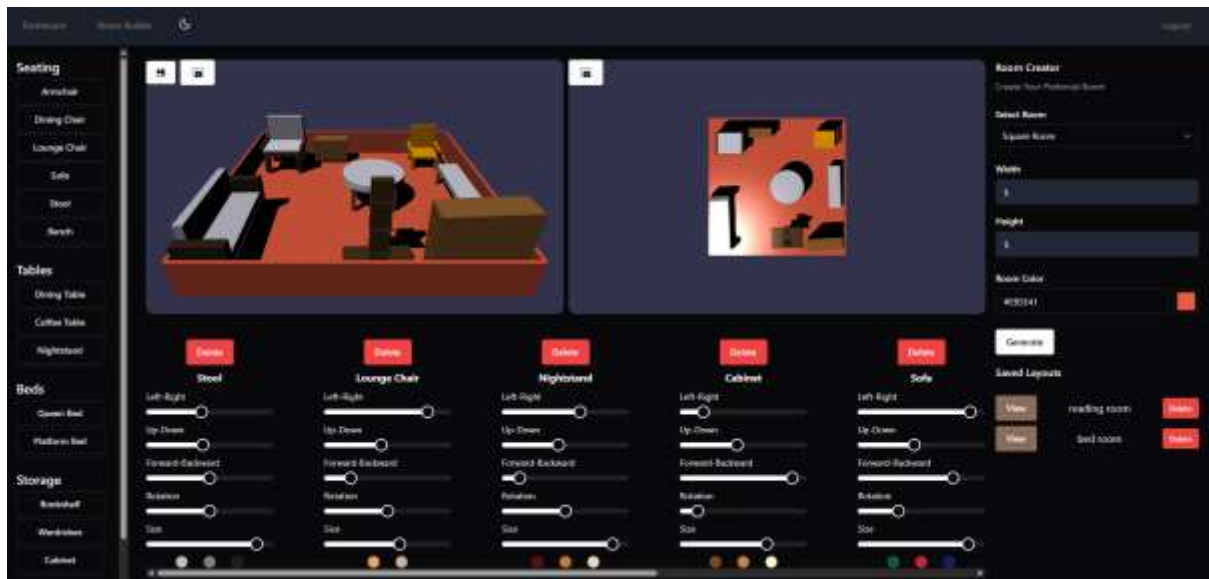


Figure 25 Room Builder Image 1

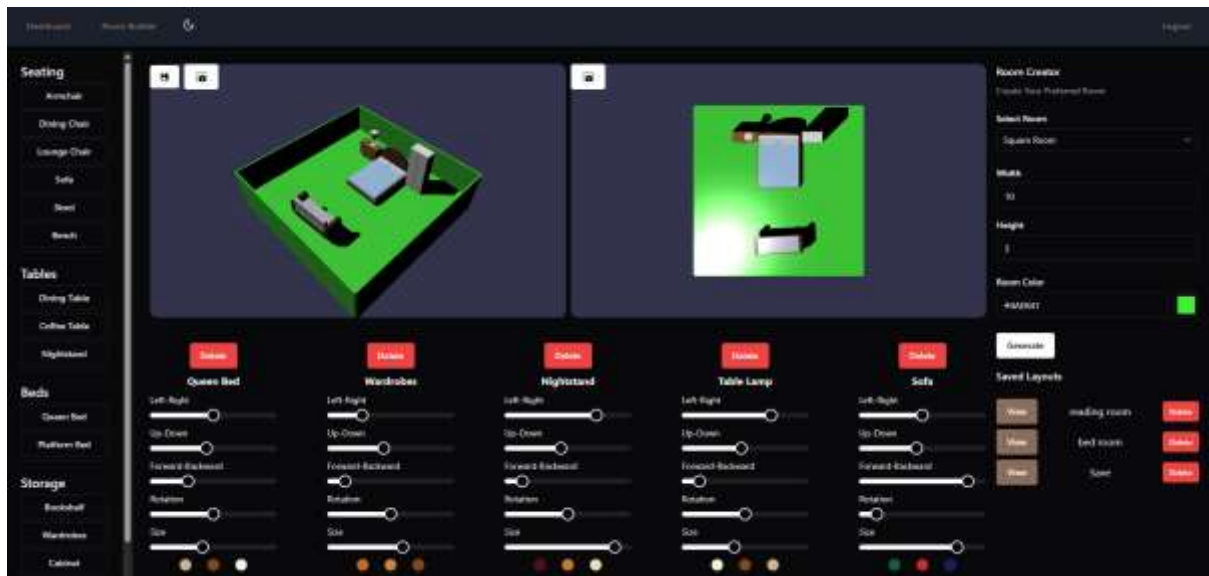


Figure 26 Room Builder Image 2